



Testing

Given a customer scenario, describe and recommend a unified test data strategy that utilizes representative data in a secure manner throughout the development lifecycle.

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After studying this topic, you should be able to:

-  Define a unified test data strategy and identify its elements.
-  Describe the considerations for developing a unified test data strategy for Salesforce development.
-  Identify the considerations for test data management in a unified test data strategy.
-  Recommend suitable test data strategies for the given technical requirements.



Introduction

A **unified test data strategy** is necessary to ensure the usage of **consistent, reliable, and secure test data** that is representative across all stages of the Salesforce development lifecycle.

For reliable test data generation, **data factory methods** and **scripts** can be used. Tools such as **Data Loader** can be used for bulk data operations in non-production environments. **Salesforce Data Mask** can be used to automatically mask sensitive production data in sandbox environments. **Mock data** that mimics production data shape and volume can be generated.

Batch Apex can be used to delete test data on a regular basis. **Automated test data loading** can be used for reliable results and when no human oversight is required. Multiple related records can be inserted in a single call or transaction using **REST API composite requests** or **sObject tree requests**.



Unified Test Data Strategy

Representative & Secure Test Data

Test data should adhere to the regulations and accurately reflect the production environment's complexity and variability without exposing sensitive information.

Test Data Creation

Representative test data can be created by using data factory methods and scripts or using tools such as Bulk API for bulk data operations.



Data Masking

Salesforce Data Mask can be utilized to automatically mask sensitive production data in sandbox environments to comply with regulations.

Sandboxes & Data Refresh

Different sandbox types can be utilized to test with representative data at various stages of the development lifecycle.

Considerations

To delete a large volume of test data, a batch Apex job can be used. Automated test data loading can be used for reliable results. REST API composite requests and sObject tree requests can be used for inserting related records.

Unified Test Data Strategy

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Unified Test Data Strategy

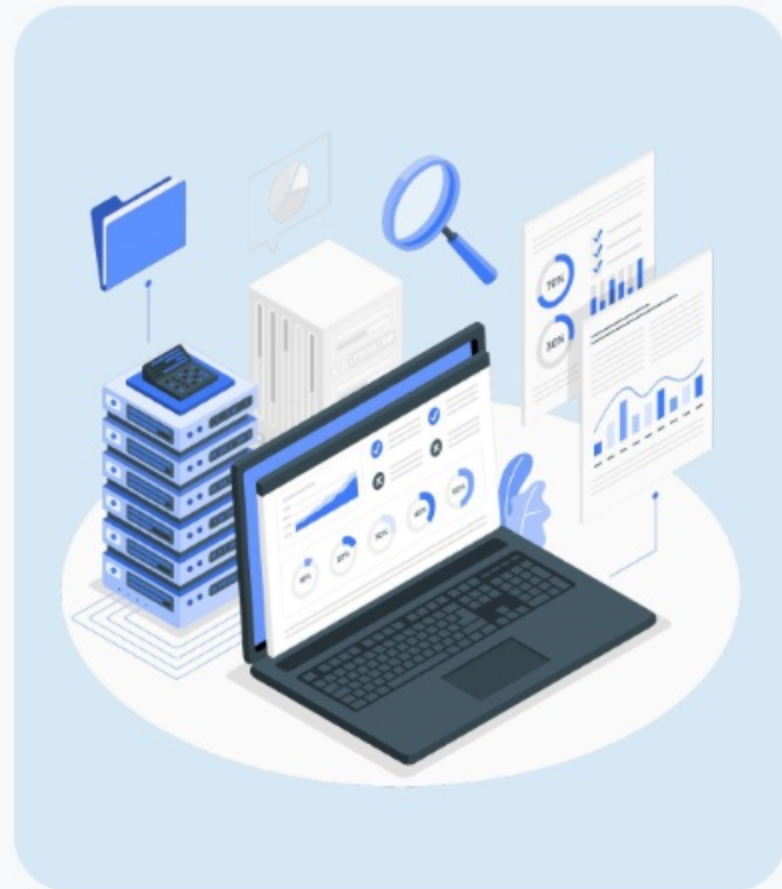
A unified test data strategy ensures **consistent, reliable, and secure test data** is used across all stages of the development lifecycle, from development to testing, and staging to production.

❖ REPRESENTATIVE DATA

Test data should **accurately** reflect the **production environment's complexity** and **variability** without exposing **sensitive information**.

❖ SECURITY & COMPLIANCE

Test data should adhere to **data protection regulations (e.g., GDPR, HIPAA, etc.)**. **Data masking, anonymization, or pseudonymization** techniques should be implemented to secure sensitive and personally identifiable information (PII).



Developing a Unified Test Data Strategy

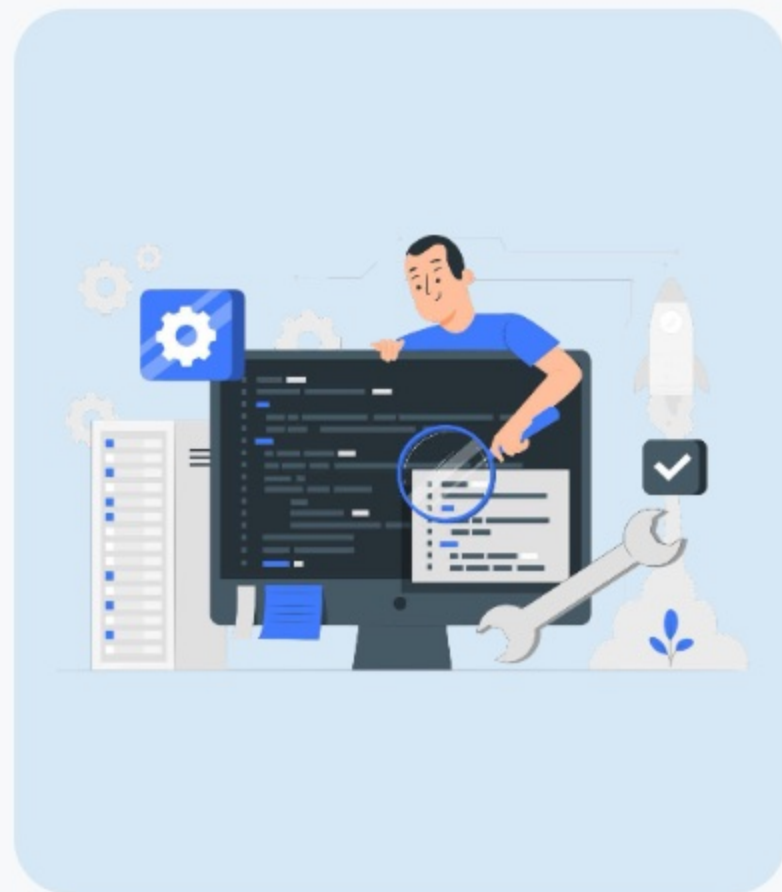
For the development of a unified test data strategy, **clear guidelines** should be established for **test data management**, including **test data creation**, **storage**, **usage**, and **disposal policies**.

❖ TEST DATA CREATION

Data factory methods and scripts can be used to generate test data that mimics real-world scenarios. Tools such as **Data Loader** can be used for **bulk data operations** in non-production environments.

❖ DATA MASKING

Data masking techniques can be implemented to **obscure sensitive data** while maintaining its usability for testing purposes. **Salesforce Data Mask** can automatically mask data in sandbox environments.



Developing a Unified Test Data Strategy

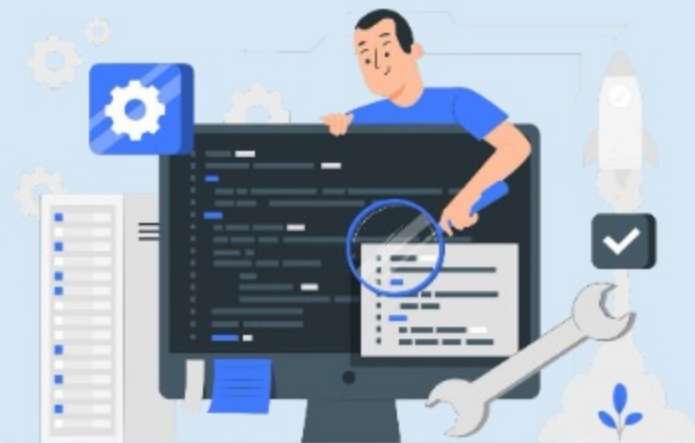
For the development of a unified test data strategy, **clear guidelines** should be established for **test data management**, including **test data creation**, **storage**, **usage**, and **disposal policies**.

❖ MOCK DATA

If regulations prohibit the use of production data for testing, **mock data** that **mimics production data shape and volume** can be generated.

❖ SANDBOX ENVIRONMENTS

Different types of sandboxes (Developer, Developer Pro, Partial Copy, and Full) can be leveraged to **manage and test with representative data** at various stages.



Developing a Unified Test Data Strategy

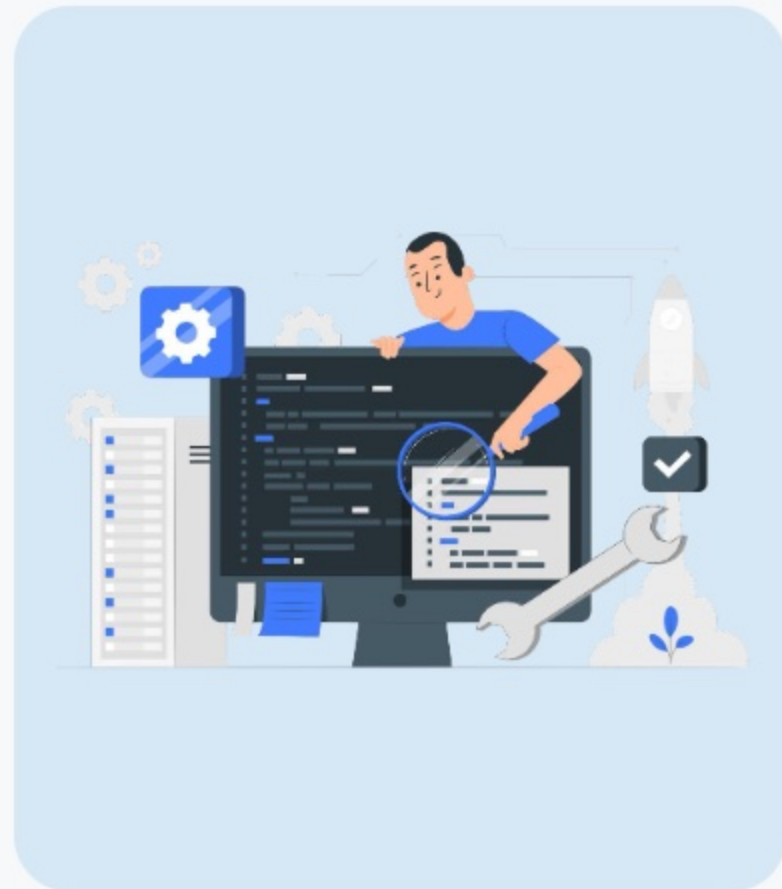
For the development of a unified test data strategy, **clear guidelines** should be established for **test data management**, including **test data creation**, **storage**, **usage**, and **disposal policies**.

❖ DATA REFRESH

Regularly **refreshing test data** in sandbox environments from production ensures that it remains representative. **Sandbox refresh strategies** can be used to manage **how** and **when** data is updated in test environments.

❖ MONITORING & AUDITING

Monitoring should be implemented to **track usage** and **access to test data**, ensuring **compliance with data policies**. **Periodic audits** should be conducted to validate adherence to the **test data strategy** and **data protection regulations**.



Considerations for Test Data Management

Certain **considerations** are applicable for the management of **test data in a unified test data strategy**.



TEST DATA DELETION

If test data needs to be **deleted** on a regular basis, an **Apex batch job** can be utilized.



AUTOMATION

Automated test data loading can be used for **reliable** results and when **no human oversight** is required.



RELATED DATA

To load **multiple related records** in a single call or transaction, **REST API composite request** or **sObject tree request** can be utilized.

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Wellness, a rapidly expanding healthcare solutions provider, leverages Salesforce to manage patient records, appointments, and interactions across its global network of clinics and hospitals. As part of its commitment to provide exceptional patient care and ensure data privacy, the company faces the challenge of testing new Salesforce features and customizations with data that accurately reflects real-world use cases without compromising patient confidentiality.

Given the sensitive nature of healthcare data, including personal health information (PHI) that must comply with strict regulatory standards like HIPAA in the U.S. and GDPR in Europe, the company needs a strategic approach to manage test data in a representative and secure manner across its development lifecycle. The company's architect needs to recommend appropriate strategies for requirements related to the same.

Requirements & Solutions

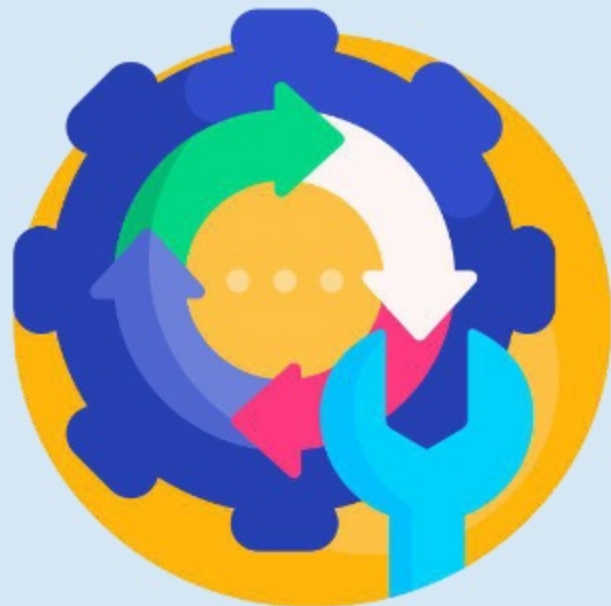
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Requirement 1



SCENARIO

A Salesforce development team of the company is working on a large customization project and is in the process of refining its testing strategies. It would like to introduce large data volume testing using representative data to resolve performance defects earlier in the development lifecycle. However, regulations prohibit the use of sensitive production data in non-production environments. Furthermore, the team wants a suitable approach to delete 2 GB of test data in a partial copy sandbox every night, setting the sandbox to a fresh state for the next day's testing.

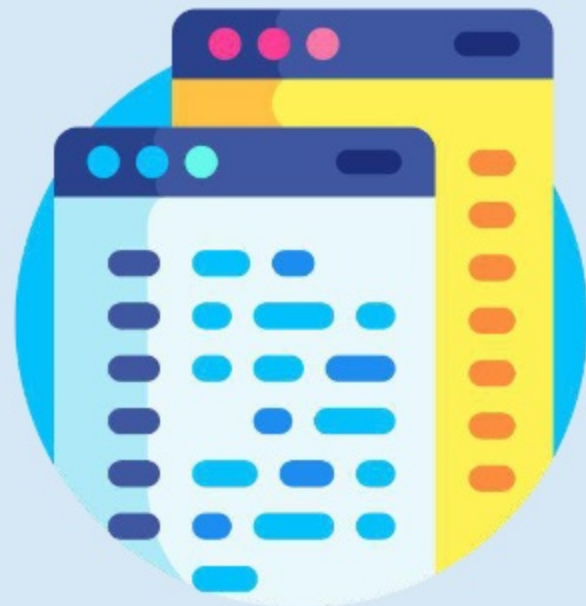


Solution 1



SOLUTION

The development team can utilize **Salesforce Data Mask** to automatically mask sensitive production data in sandbox environments. Another option is **generating and loading mock data** that mimics production data shape and volume. To delete the test data in the partial copy sandbox every night, a **batch Apex job** can be executed at a specific time.



Requirement 1



SCENARIO

A small team of Salesforce developers is working on a custom application for managing patient information. Various features of the application are already in the testing stage. The team wants a suitable strategy to insert test data that represents records of multiple related objects that mimic real patient interactions, medical records, and appointment scheduling. In addition, the team is considering automated test data loads and would like the architect's assistance in choosing automated test data loading over manual loading.



Solution 1



SOLUTION

For the loading of test data that represents records of multiple related objects, **REST API composite requests** or **REST API sObject tree requests** can be utilized. A composite request is a single POST request that represents a series of REST API requests. An sObject tree request comprises a collection of nested, parent-child records with a single root record.

Automated test data loading is useful for **reliable results** and when **no human oversight** is required for loading test data.



Learn More

- [!\[\]\(e4d819883f134098bdcdfd1ccb861584_img.jpg\) Secure Your Sandbox Data with Salesforce Data Mask](#)
- [!\[\]\(454907c4a63da18a91379af2a30f661a_img.jpg\) Common Test Utility Classes for Test Data Creation](#)
- [!\[\]\(d994b90baa0188fb29f9c0fc0166055d_img.jpg\) Batch Apex](#)
- [!\[\]\(99c94d76b4eeac64b913b566e86c3373_img.jpg\) Composite](#)
- [!\[\]\(d8883b50654506e853f29c970af5fcde_img.jpg\) sObject Tree](#)

